Bootcamp Infosheet

Instructor

Name: *Dr. Veronica Red*

Bio: *Dr. Veronica Red (they / them) received their Ph.D. in Biomathematics with a Minor in Statistics from North Carolina State University. They have a certification to teach Deep Learning courses from Nvidia, and they have experience teaching everything from Intro to R/Python to Conversational AI Bots. Dr. Red teaches Machine Learning courses for Data Society to professionals at the State Department, Comcast, and many other organizations. Dr. Red also has many years of experience working as a data scientist for government projects (statistical analysis, data visualization, network analysis, machine learning) and private sector (machine learning, data visualization, network analysis, statistical analysis, and predictive modeling).*

Bootcamp Details

Bootcamp Title: *LangGraph-Powered AI Agents for Rapid Business Impact*

Number of Days: *4*

Hours per Day: *3*

Type of Instruction: *Lecture, chat, polling questions, and participant guide activities.*

Description: *This intensive, four-day bootcamp transitions participants from foundational AI concepts to the cutting edge of Agentic AI orchestration using the LangGraphframework. Learners will begin by deconstructing agent architectures—including ReAct and MRKL paradigms—and mastering the theory that underpins modern data science. The curriculum then dives deep into the LangGraph framework,* *teaching students how to design stateful, cyclical workflows that overcome the limitations of linear chains. By the final day, participants will be able to describe production-grade agents equipped with Human-in-the-Loop (HITL) checkpoints, short- and long-term memory modules, and specialized tool-integration for RAG (Retrieval-Augmented Generation) use cases.*

NOTE: This course will be taught through instructor demos and non-coding activities. High proficiency in Python is still expected to participate.

Target Audience: *This course is designed for data scientists, AI engineers, and technical product managers who have experience with basic LLM prompting and are ready to build complex, autonomous, or semi-autonomous AI systems. It is ideal for those looking to move beyond simple chat interfaces into sophisticated, state-managed agent orchestration.*

Technologies: *This course focuses on the conceptual architecture and strategic design patterns of the LangGraph and LangChain ecosystems. While the instructor will demonstrate these concepts using code-based visualizations and live logic-mapping, the webinar is designed for high-level comprehension rather than manual configuration. We will explore how these orchestration frameworks interact with state-of-the-art models and specialized search tools, focusing on the flow of data, state management, and system constraints.*

Prerequisites: *This course is accessible to professionals with a foundational understanding of Generative AI and Large Language Models (LLMs). Learners should be familiar with basic Python objects, AI terminology, such as prompting, context windows, and hallucinations, and have a conceptual interest in how complex business processes can be mapped into automated workflows. An appreciation for logical flowcharts or process mapping will be highly beneficial for the architectural design portions of the curriculum.*

Student References: *Class slides and participant guide.*

Bootcamp Syllabus

Day 1: Contextual grounding for LangGraph and AI Agents

- **Agent architecture and design patterns**
    - **Define key terminology: user request, agent/brain, planning, and memory**
    - **Differentiate between common AI Agent types and their functions**
- **Design outcomes for AI Agents**
    - **Match Agentic AI paradigms to use cases**
    - **Compare ReACT, MRKL, and HiTL paradigms within Agentic AI scenarios**
- **Review Graph components**
    - **Introduce networks and their use in data science**
    - **Discuss networks use cases**
    - **Define a network graph and its components**

Day 2: Foundations of LangGraph

- **Overview of tools and frameworks for AI Agents**

    - Introduce LangGraph in the context of AI agent orchestration
    - Compare LangGraph conceptually to LangChain and other frameworks
    - Identify use cases where graph-based workflows excel (cyclical workflows, branching logic)
    - Discuss trade-offs: modularity, debugging, memory management, and control
- LangGraph end-to-end workflow
    - Describe the end-to-end process for creating a simple chatbot using LangGraph
- LangGraph components
    - Describe core elements: state, nodes, edges and types
    - Identify key details of core elements (state, nodes, edges)
    - Design the workflow for a decision-maker in LangGraph

Day 3: LangGraph LLM workflow and ReACT agent fundamentals

- Devise a simple LLM workflow in Langraph
    - Discuss messages and chat models for conversation State
    - Design a stateful chatbot Agent using LangGraph
    - Visualize a chatbot workflow with Graph diagrams
- ReACT Agents in LangGraph
    - Describe ReAct principles and best practices
    - Design ReAct nodes and edges (looping and conditional)
    - Visualize a simple ReAct agent

Day 4: Components of ReACT Agents in LangGraph

- Human-in-the-Loop
    - Define best practices and use cases for HITL
    - Describe the interrelationship between ReAct and HITL
    - Identify locations for HITL checkpoints in ReACT agents
- Tool-using ReAct agent
    - Review types of tools that an agent might use
    - Introduce use case for creating a ReAct agent that uses tools
    - Identify the relevant tools for a ReAct Agent use case (RAG)
    - Design a Graph for a ReAct agent using RAG as a tool
- Agents with Memory
    - Review memory modules available to LangGraph (short-term and long-term)
    - Introduce use case for memory-holding agent